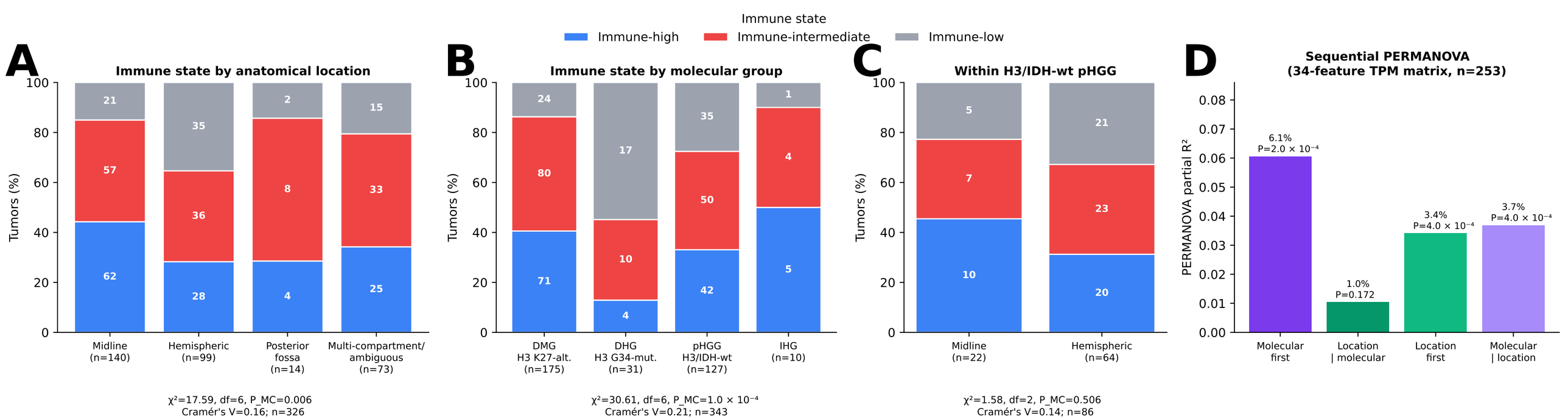

Immune state
Immune-high
Immune-intermediate
Immune-low
A
Immune state by anatomical location
Tumors (%)
Midline (n=140)
Hemispheric (n=99)
Posterior fossa (n=14)
Multi-compartment/ ambiguous (n=73)
χ²=17.59, df=6, P_MC=0.006
Cramér's V=0.16; n=326
B
Immune state by molecular group
DMG H3 K27-alt. (n=175)
DHG H3 G34-mut. (n=31)
pHGG H3/IDH-wt (n=127)
IHG (n=10)
χ²=30.61, df=6, P_MC=1.0 × 10⁻⁴
Cramér's V=0.21; n=343
C
Within H3/IDH-wt pHGG
Midline (n=22)
Hemispheric (n=64)
χ²=1.58, df=2, P_MC=0.506
Cramér's V=0.14; n=86
D
Sequential PERMANOVA
(34-feature TPM matrix, n=253)
PERMANOVA partial R²
6.1%
P=2.0 × 10⁻⁴
1.0%
P=0.172
3.4%
P=4.0 × 10⁻⁴
3.7%
P=4.0 × 10⁻⁴
Molecular first
Location | molecular
Location first
Molecular | location